The Effects of the Basel III Financial Regulatory Framework on Bank Lending: Evidence from U.S. Banks

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Abstract

This paper examines how financial regulatory reforms, specifically Basel III, affect bank lending behavior in the United States. Using a large panel dataset of U.S. commercial banks spanning several decades, I construct a measure of regulatory exposure based on pre-regulation capital ratios. I employ a difference-in-differences (DiD) framework with bank and time fixed effects (FE), complemented by an event study design and debiased machine learning (DML) methods to control for high-dimensional confounders.

Contrary to the standard prediction that higher capital requirements reduce lending, I find that banks more exposed to Basel III increase lending relative to less exposed banks following the policy implementation. The event study analysis supports the assumption of identifying parallel trends after accounting for bank-specific trends. These findings suggest that banks adjust to regulatory constraints along multiple margins, including capital structure and asset composition, rather than uniformly reducing lending. The results show the importance of heterogeneous responses in evaluating the effects of financial regulation.

I would like to thank Professor Thomas Russell for all his helpful comments. All errors are my own.

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1 Introduction and Motivation

Financial regulation plays a central role in maintaining the stability of the banking system. Banks are inherently fragile institutions due to their high leverage and maturity transformation, making them vulnerable to systemic crises. The 2008 global financial crisis highlighted these vulnerabilities and led to the introduction of Basel III, comprehensive regulatory reforms designed to strengthen capital requirements and improve the resilience of financial institutions.

A key question in both academic research and policy debates is how such regulatory changes affect bank lending behavior. Standard economic theory predicts that higher capital requirements increase the cost of lending, leading banks to reduce credit supply. However, the empirical evidence is mixed, and the effects of regulation can depend on bank characteristics and the broader economic environment.

This paper investigates the impact of Basel III on bank lending behavior in the United States. Specifically, I ask: *How do regulatory changes, such as the introduction of Basel III, affect bank lending, and how do these effects vary across banks?*

To answer this question, I construct a measure of regulatory exposure based on bank pre-regulation capital ratios. Banks with lower capital ratios prior to Basel III are expected to be more affected by the new requirements. I then implement a difference-in-differences (DiD) framework comparing high-exposure and low-exposure banks before and after the policy implementation. The empirical strategy includes bank and time fixed effects to control for unobserved heterogeneity and macroeconomic shocks. In addition, I employ an event study design to validate the identifying assumptions and incorporate debiased machine learning methods to flexibly control for high-dimensional confounders.

The main finding of the paper is that, contrary to standard predictions, banks more exposed to Basel III increase lending relative to less exposed banks following the policy implementation. The event study shows no evidence of differential pre-trends after controlling for bank-specific trends, supporting a causal interpretation of the results. These findings suggest that banks adjust to regulatory constraints across multiple margins, including raising capital and reallocating assets, rather than uniformly reducing lending.

This paper contributes to the literature by providing new evidence on the heterogeneous effects of capital regulation using a large panel dataset and modern causal inference techniques. The results have important implications for the design of financial regulation and the evaluation of its

effects on credit supply.

2 Literature Review

This paper contributes to a large body of literature examining the relationship between bank capital regulation and lending behavior. Following the global financial crisis of 2008, regulatory frameworks such as Basel III were introduced to strengthen bank resilience through higher capital and liquidity requirements. A central question in this literature is whether tighter capital regulation constrains credit supply.

A substantial portion of the empirical literature finds that increases in capital requirements lead to a reduction in bank lending. Using bank-level panel data, Bridges et al. (2014) analyze the impact of capital requirement changes in the United Kingdom and find that higher capital requirements are associated with a contraction in lending. Similarly, Jiménez et al. (2017), using loan-level data from Spain, show that weaker banks reduce credit supply following regulatory tightening. These studies provide strong evidence in favor of the traditional view that capital regulation restricts lending by increasing the cost of intermediation.

However, other studies emphasize that the effects of capital regulation are heterogeneous across banks and may not lead to a uniform contraction in lending. Aiyar, Calomiris, and Wieladek (2014) examine the effects of capital requirements in the UK and document substitution effects, whereby constrained banks reduce lending while less constrained institutions expand their lending activity. This highlights the importance of general equilibrium effects in banking markets, where the aggregate impact of regulation depends on how different institutions adjust.

Cross-country analyses further support the presence of heterogeneous effects. Evidence from the International Monetary Fund (2017) suggests that the impact of capital regulation on lending varies depending on the degree of capital constraints and the structure of the banking system. Similarly, OECD (2011) simulations indicate that while Basel III may reduce lending in the short run, the long-run effects are more nuanced and depend on banks' adjustment mechanisms.

More recent research has also emphasized that banks can adjust to regulatory constraints along multiple margins beyond lending quantities. Hazra et al. (2023) find that while capital regulation may reduce lending in some cases, banks simultaneously adjust pricing and portfolio composition. This suggests that focusing solely on lending quantities may provide an incomplete picture of how regulation affects bank behavior.

Despite these contributions, many gaps remain in the literature. First, many studies rely on relatively short time horizons or limited cross-sectional variation, particularly in the U.S. context. Second, identifying causal effects remains challenging due to endogeneity concerns, including the non-random assignment of regulatory constraints and the influence of macroeconomic conditions. Third, not many studies combine modern causal inference techniques with high-dimensional datasets to explore heterogeneous treatment effects.

This paper addresses these gaps in several ways. Using a large panel dataset of U.S. commercial banks spanning multiple decades, it provides a comprehensive analysis of the long-run effects of Basel III-style regulation. In particular, this paper builds on and extends the findings of Ben Naceur, Pépy, and Roulet (2017) by focusing on U.S. bank-level data and incorporating modern causal inference techniques. The empirical strategy combines a difference-in-differences framework with an exposure-based treatment definition, allowing for credible causal identification. In addition, the use of debiased machine learning methods enables flexible control for high-dimensional confounders and facilitates the analysis of heterogeneous responses.

The findings of this paper contribute to the growing literature emphasizing that the effects of financial regulation are not uniform across institutions. In contrast to the standard prediction of reduced lending, the evidence shows that more exposed banks may increase lending relative to less exposed banks following regulatory changes. This is consistent with theories of heterogeneous adjustment and policy effects, and highlights the importance of considering multiple behavioral margins when evaluating regulatory policy.

3 Data and Empirical Context

This paper uses a large panel dataset of U.S. commercial banks constructed from regulatory call reports. The data are drawn from Correia, Luck, and Verner (forthcoming), which compiles detailed quarterly balance sheet and income statement information for U.S. banks over an extended historical period. The dataset contains approximately 2.5 million bank-quarter observations spanning from 1959 to 2024, covering more than 24,000 unique institutions.

For the purposes of this analysis, I focus on the period surrounding the implementation of Basel III, which was introduced in the United States following the 2008 financial crisis and phased in beginning around 2013. The resulting dataset is a panel of bank-level observations at a quarterly frequency, allowing for the analysis of both cross-sectional and time-series variation in lending

behavior.

3.1 Variable Construction

The primary outcome variable in this study is bank lending growth. Specifically, I construct loan growth as the log difference in total loans between consecutive quarters:

$$LoanGrowth_{it} = 100 \times (\log(Loans_{it}) - \log(Loans_{it-1}))$$

This measure captures percentage changes in lending activity and is commonly used in the empirical banking literature.

The key explanatory variable is a measure of regulatory exposure based on banks pre-regulation capital positions. I construct the capital ratio as equity divided by total assets:

$$CapitalRatio_{it} = \frac{Equity_{it}}{Assets_{it}}$$

To capture exogenous variation in regulatory exposure, I compute the average capital ratio for each bank over a pre-treatment period (2008–2010). While equity-to-assets is not identical to regulatory capital ratios used in Basel III, it provides a consistent and widely available proxy across the full sample. Banks with lower pre-period capital ratios are considered more exposed to Basel III, as they are required to make larger adjustments to meet the new regulatory standards. I define both a continuous exposure measure and a binary indicator for high exposure based on the median of the pre-period capital ratio distribution.

The treatment variable is defined as an indicator for the post-Basel III period:

$$Post_t = \begin{cases} 1 & \text{if } t \geq 2013 \\ 0 & \text{otherwise} \end{cases}$$

In addition to the main variables of interest, I include several control variables to account for bank characteristics and risk. These include the logarithm of total assets as a measure of bank size, the ratio of loan loss reserves to total loans as a proxy for risk, deposit growth, and measures of funding structure such as the share of demand and time deposits. Profitability is captured using return on assets.

3.2 Summary Statistics and Correlations

Table 1 presents summary statistics for the main variables used in the analysis. The average quarterly loan growth rate is approximately 1.29 percent, with substantial variation across banks and over time. Banks also exhibit significant heterogeneity in size and capitalization, with the average equity-to-assets ratio equal to 0.11.

Table 1: Summary Statistics

Variable	Mean	Std. Dev.	Min	Max
Loan Growth	1.29	5.07	-12.76	22.97
Log Assets	12.35	1.36	9.74	17.08
Equity / Assets	0.11	0.05	-2.15	1.00
LLR / Loans	0.02	0.01	0.00	0.05
Deposit Growth	1.48	5.35	-11.78	24.22

The data reveal substantial heterogeneity across banks, particularly in terms of size and capitalization. The relatively low pairwise correlations between key variables further suggest that multicollinearity is not a major concern in the empirical analysis.

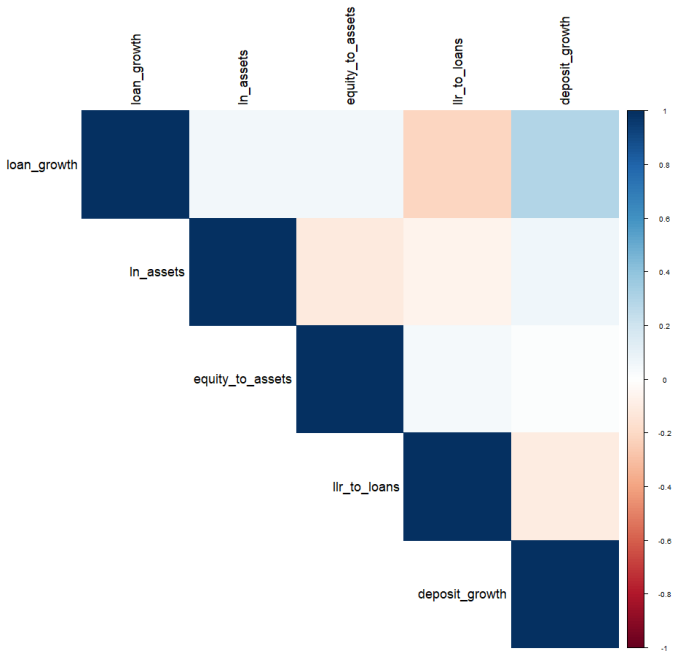


Figure 1: Correlation matrix of key variables

Figure 1 presents the correlation matrix of key variables. While lending is positively correlated with deposit growth and negatively correlated with risk, capital ratios exhibit only weak correlations with lending outcomes. This highlights the need for a causal identification strategy.

3.3 Empirical Context

Basel III was introduced in response to the global financial crisis with the goal of strengthening bank resilience through higher capital and liquidity requirements. In the United States, implementation occurred gradually beginning in the early 2010s, with major provisions taking effect around 2013.

A key feature of Basel III is that its impact is not uniform across banks. Institutions with lower pre-existing capital ratios are more affected, as they must increase equity relative to assets to meet regulatory standards. This heterogeneity provides the basis for the empirical strategy employed in this paper.

The combination of a large panel dataset, rich balance sheet information, and variation in regulatory exposure allows for a detailed analysis of how banks adjust their lending behavior in response to changes in capital regulation.

4 Methodology

This paper estimates the causal effect of Basel III on bank lending using a difference-in-differences (DiD) framework that exploits cross-sectional variation in banks' exposure to the regulation. The empirical strategy compares changes in lending behavior across banks with different pre-regulation capital positions before and after the implementation of Basel III.

4.1 Baseline Specification

The baseline empirical model is given by:

$$Y_{it} = \beta(Post_t \times Exposure_i) + \alpha_i + \gamma_t + X'_{it}\delta + \epsilon_{it}$$

where Y_{it} denotes the lending outcome for bank i at time t, measured as the growth rate of loans. $Post_t$ is an indicator variable equal to one for periods following the implementation of Basel III (beginning in 2013), and zero otherwise. $Exposure_i$ is a measure of regulatory exposure based on the bank's pre-regulation capital ratio.

The coefficient of interest, β, captures the differential change in lending for more exposed banks relative to less exposed banks after the introduction of Basel III. A negative value of β would be consistent with the standard prediction that tighter capital requirements reduce lending, while a positive value would suggest alternative adjustment mechanisms.

The model includes bank fixed effects, α_i, which control for time-invariant differences across banks, such as business models or risk preferences. Time fixed effects, γ_t, absorb common macroeconomic shocks affecting all banks in a given period, including business cycle fluctuations and financial crises. The vector X_{it} includes time-varying control variables such as bank size, risk measures, funding structure, and profitability.

4.2 Exposure Definition

A key component of the identification strategy is the construction of the exposure variable. I define exposure using the bank's average equity to assets ratio over a pre-treatment period (2008–2010). Banks with lower capital ratios are considered more exposed to Basel III, as they are required to make larger adjustments to meet the new regulatory standards.

To facilitate interpretation, I construct both a continuous and a binary measure of exposure. The continuous measure is defined as the negative of the pre-period capital ratio, such that higher values correspond to greater exposure. The binary measure classifies banks as high exposure if their pre-period capital ratio falls below the sample median.

4.3 Identification Strategy

The identification strategy relies on comparing lending behavior across banks with different levels of exposure before and after Basel III. The key identifying assumption is the parallel trends assumption, which requires that, in the absence of the policy change, lending trends for high-exposure and low-exposure banks would have evolved similarly over time.

Bank fixed effects ensure that identification comes from within-bank variation over time, while time fixed effects control for aggregate shocks. The interaction term $Post_t \times Exposure_i$ isolates the differential effect of Basel III on more exposed banks.

Figure 2 illustrates the causal structure of the empirical strategy. Bank characteristics and macroeconomic conditions affect both regulatory exposure and lending behavior. The inclusion of bank and time fixed effects helps control for these confounding factors



Figure 2: Directed Acyclic Graph (DAG)

4.4 Event Study Specification

To assess the validity of the parallel trends assumption and examine the dynamic effects of the policy, I estimate an event study specification:

$$Y_{it} = \sum_{k \neq -1} \beta_k \left(D_{t=k} \times Exposure_i \right) + \alpha_i + \gamma_t + X_{it}' \delta + \epsilon_{it}$$

where $D_{t=k}$ is an indicator for event time relative to the implementation of Basel III, and the period immediately prior to implementation ($k = -1$) is omitted as the reference category. The coefficients β_k trace the evolution of the treatment effect over time.

This specification allows for a direct test of the parallel trends assumption by examining whether pre-treatment coefficients are statistically indistinguishable from zero. It also provides insight into the timing and persistence of the policy's effects on lending.

4.5 Bank-Specific Trends

To further address concerns about differential pre-trends across banks, I augment the baseline specification by including bank-specific linear time trends. This allows each bank to follow its own underlying trajectory in lending behavior, reducing the risk that estimated treatment effects are driven by pre-existing differences in trends across banks.

4.6 Debiased Machine Learning

In addition to the fixed effects framework, I implement a debiased machine learning (DML) approach to flexibly control for high-dimensional confounders. Specifically, I use machine learning methods to partial out the effects of control variables on both the outcome and treatment variables, and then estimate the causal parameter using residualized variables.

This approach allows for nonlinear relationships and interactions among covariates without imposing strong parametric assumptions, while preserving valid statistical inference. The DML framework provides a robustness check for the main DiD results and helps ensure that the estimated effects are not driven by model misspecification.

4.7 Discussion of Identification

The empirical strategy addresses several key threats to identification. Time fixed effects control for macroeconomic shocks, such as the financial crisis and business cycle fluctuations, while bank fixed

effects account for persistent differences across institutions. The event study provides evidence on the validity of the parallel trends assumption, and the inclusion of bank-specific trends further mitigates concerns about differential pre-existing trajectories.

Nevertheless, some limitations remain. Basel III was implemented gradually rather than as a single discrete shock, and exposure is measured using a proxy for regulatory capital rather than exact regulatory thresholds. These considerations are taken into account when interpreting the results.

5 Results

5.1 Baseline Difference-in-Differences Estimates

Table 2: Effect of Basel III on Bank Lending

	(1) Binary Exposure	(2) Continuous Exposure
Post × Exposure	0.654***	14.265***
	(0.055)	(1.967)
Log Assets	-0.704***	-0.579***
	(0.094)	(0.088)
LLR / Loans	-189.07***	-187.47***
	(4.283)	(4.400)
Demand Deposit Share	-0.385	-0.295
	(0.264)	(0.261)
Time Deposit Share	-1.917***	-1.513***
	(0.391)	(0.399)
ROA	-24.36***	-21.10***
	(4.417)	(4.176)
Observations	371,507	371,507
Bank FE	Yes	Yes
Time FE	Yes	Yes
Clustered SE	Bank	Bank

Standard errors in parentheses.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 2 presents the main difference-in-differences estimates of the effect of Basel III on bank lending. The sample consists of 371,507 bank-quarter observations across approximately 7,600 banks over 67 quarters.

The coefficient on the interaction term $Post_t \times HighExposure_i$ is positive and statistically significant at the 1% level. The estimated effect is 0.65 percentage points (standard error = 0.055),

indicating that banks with lower pre-regulation capital ratios increase lending relative to less exposed banks following the implementation of Basel III.

This result is economically meaningful given the average level of loan growth in the sample. The positive sign contrasts with the standard prediction that tighter capital requirements reduce lending, and instead suggests that banks adjust along multiple margins rather than shrinking credit supply.

The results are robust to alternative specifications. Column (2) reports estimates using a continuous measure of exposure, and yields a positive and statistically significant coefficient, confirming that the main findings are not sensitive to the definition of regulatory exposure.

These findings have a statistical significance level of 1% and are economically meaningful given the average level of loan growth in the sample. The magnitude suggests that regulatory exposure is correlates with a change in lending behavior.

The signs of the control variables are mostly consistent with expectations. Larger banks exhibit lower loan growth, while higher loan loss reserves are associated with reduced lending, reflecting the role of risk. Measures of funding structure and profitability also influence lending behavior, though not all coefficients are statistically significant.

Importantly, the main effects of $Post_t$ and $HighExposure_i$ are absorbed by time and bank fixed effects, respectively, ensuring that identification comes from within-bank variation over time.

5.2 Continuous Exposure Specification

To assess the robustness of the results, I estimate the model using a continuous measure of exposure based on the pre-period capital ratio. The coefficient on the interaction term $Post_t \times Exposure_i$ is positive and highly statistically significant.

Specifically, the estimated coefficient is approximately 14.26 (standard error = 1.97), indicating that greater exposure to Basel III is associated with higher loan growth following implementation. The positive sign is consistent with the baseline binary specification, confirming that the results are not sensitive to how exposure is defined.

These findings suggest that the relationship between regulatory exposure and lending is monotonic and robust across alternative specifications.

5.3 Event Study Analysis

To evaluate the validity of the identifying assumptions and examine the dynamic effects of Basel III, I estimate an event study specification with bank-specific trends.

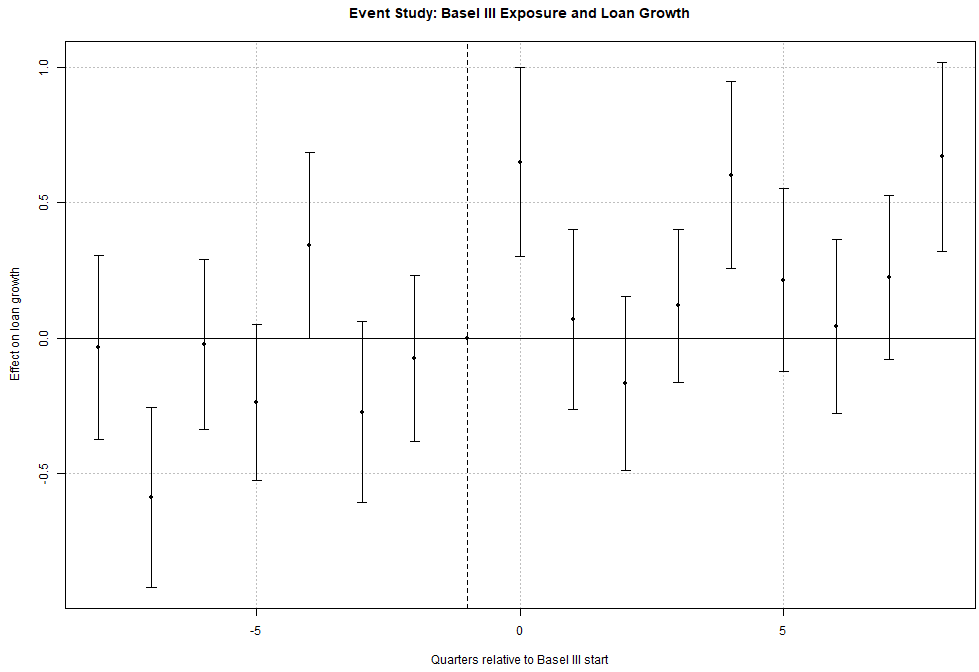


Figure 3: Event Study: Dynamic Effects of Basel III

The event study results show no systematic evidence of differential pre-trends between high- and low-exposure banks. The estimated coefficients in the pre-treatment period are generally small and statistically indistinguishable from zero, supporting the parallel trends assumption underlying the difference-in-differences framework.

In the post-treatment period, the coefficients become positive and statistically significant in several periods, indicating a divergence in lending behavior following the implementation of Basel III. The effect appears to emerge gradually over time, consistent with the phased implementation of the regulation and the banking adjustment process.

5.4 Debiased Machine Learning Results

To assess the robustness of the results and incorporate modern data-driven methods, I implement a debiased machine learning (DML) approach. This framework uses machine learning algorithms to flexibly control for high-dimensional confounders and allows for nonlinear relationships between co-

variates and the outcome. The machine learning component of the DML procedure is implemented using gradient boosting methods, which allow for flexible modeling of nonlinear relationships and interactions among covariates.

The debiased machine learning (DML) framework can be represented as:

$$Y_{it} = \theta D_{it} + g(X_{it}) + \epsilon_{it}$$

where Y_{it} is the outcome variable, D_{it} is the treatment variable, and X_{it} is a high-dimensional set of control variables. The function $g(\cdot)$ captures potentially complex and nonlinear relationships between covariates and the outcome.

The DML approach proceeds by first estimating the nuisance functions using machine learning methods. Specifically, the outcome and treatment are residualized as follows:

$$\tilde{Y}_{it} = Y_{it} - \hat{g}(X_{it})$$
$$\tilde{D}_{it} = D_{it} - \hat{m}(X_{it})$$

The causal parameter θ is then estimated by regressing the residualized outcome on the residualized treatment:

$$\tilde{Y}_{it} = \theta \tilde{D}_{it} + u_{it}$$

Table 3: Debiased Machine Learning Estimates

	DML Estimate
Post × Exposure	-1.389***
	(0.343)

Standard errors in parentheses.
*** $p < 0.01$

The estimated treatment effect is negative and statistically significant, with a coefficient of approximately -1.39 (standard error = 0.34).

This result differs from the baseline difference-in-differences estimate, which is positive. The divergence suggests that the estimated effect of Basel III on lending may be sensitive to modeling assumptions and the way control variables are incorporated.

One possible explanation is that the DML framework captures nonlinear relationships and interactions among covariates that are not fully accounted for in the linear DiD specification. Alter-

natively, the difference may reflect the role of functional form assumptions or residual confounding in the baseline model.

To summarize, these findings highlight the importance of considering multiple empirical approaches when evaluating the effects of financial regulation.

5.5 Interpretation of Results

The baseline difference-in-differences estimates indicate that banks more exposed to Basel III increase lending relative to less exposed banks following the policy implementation. This finding contrasts with the standard prediction that tighter capital requirements reduce lending by increasing the cost of intermediation.

One possible explanation for this result is that banks adjust to regulatory constraints along multiple margins. Rather than reducing lending, more exposed banks may increase capital through retained earnings or equity issuance, allowing them to maintain or expand lending activity. In addition, general equilibrium effects may lead to a reallocation of lending across institutions, as banks respond differently to regulatory changes depending on their initial conditions.

However, the debiased machine learning (DML) estimates yield a negative and statistically significant effect of regulatory exposure on lending, consistent with the traditional view that higher capital requirements constrain credit supply. The divergence between the DiD and DML results suggests that the estimated effects are sensitive to modeling assumptions and the treatment of control variables.

One interpretation is that the linear DiD specification may not fully capture nonlinear relationships or interactions among covariates, which are more flexibly modeled in the DML framework. Alternatively, the difference may reflect the presence of heterogeneous treatment effects, where the impact of regulation varies across banks in ways that are captured differently by the two approaches.

Taken together, the results highlight the complexity of assessing the effects of financial regulation. While the DiD estimates suggest heterogeneous and potentially expansionary responses among more exposed banks, the DML results emphasize the possibility of contractionary effects once more flexible controls are incorporated. This underscores the importance of using multiple empirical approaches and interpreting results with caution.

5.6 Limitations

While the results provide evidence on the effects of Basel III on bank lending, several limitations should be noted.

First, the measure of regulatory exposure is based on the equity-to-assets ratio, which serves as a proxy for regulatory capital but does not fully capture the complexity of Basel III requirements. In particular, Basel III relies on risk-weighted capital measures and additional regulatory buffers that are not directly observed in the data.

Second, the implementation of Basel III is gradual rather than a single discrete policy shock. This may introduce measurement challenges in defining the exact timing of treatment and could attenuate estimated effects.

Third, although the empirical strategy incorporates bank and time fixed effects, as well as a rich set of controls, residual endogeneity may remain. In particular, banks may be differentially exposed to macroeconomic or regional shocks that are not fully captured by the model.

Finally, the results exhibit sensitivity to modeling choices. While the difference-in-differences estimates suggest a positive effect of regulatory exposure on lending, the debiased machine learning approach yields a negative effect. This divergence indicates that the estimated impact of Basel III may depend on functional form assumptions and the treatment of high-dimensional controls. As a result, the findings should be interpreted with caution.

These limitations highlight the challenges of identifying causal effects in complex financial systems and underscore the importance of using multiple empirical approaches.

6 Conclusion

This paper examines the impact of Basel III on bank lending behavior using a large panel of U.S. commercial banks and a difference-in-differences framework based on pre-regulation capital exposure. The results indicate that banks with lower pre-regulation capital ratios increase lending relative to less exposed banks following the implementation of Basel III. This finding contrasts with the standard prediction that tighter capital requirements reduce lending.

To assess the robustness of these findings, I implement a debiased machine learning (DML) approach that flexibly controls for high-dimensional confounders. In contrast to the baseline results, the DML estimates suggest a negative effect of regulatory exposure on lending, consistent with traditional theoretical predictions. The divergence between these approaches highlights the

sensitivity of estimated effects to modeling assumptions and underscores the importance of using multiple empirical methods.

The results suggest that the effects of financial regulation are more complex than a simple contraction in credit supply. Banks appear to adjust along multiple margins, and regulatory changes may lead to heterogeneous responses and reallocation of lending across institutions rather than a uniform decline.

These findings have important implications for policymakers. While capital regulation is designed to enhance financial stability, its effects on credit supply may vary across institutions and depend on how banks adjust their balance sheets. Understanding these heterogeneous responses is critical for designing effective regulatory policies that balance stability and credit availability.

Several questions for future research remain. First, future work could use more precise measures of regulatory capital, such as risk weighted capital ratios, to better capture the mechanisms underlying Basel III. Second, exploring regional or sectoral heterogeneity could provide insight into how regulation affects different segments of the economy. Third, further research could focus on disentangling credit supply and demand effects, which remains a central challenge in empirical banking studies. Finally, extending the analysis using alternative machine learning methods or causal inference frameworks could help reconcile differences across empirical approaches and improve understanding of regulatory impacts.

In conclusion, this paper highlights the importance of combining traditional econometric techniques with modern data science methods when evaluating policy interventions. The results contribute to a growing study emphasizing that the effects of financial regulation are heterogeneous, context dependent, and sensitive to modeling choices.

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7 Appendix

7.1 Fixed effects regression output

Dependent Variable:	loan_growth	
Model:	(1)	(2)
Variables		
ln_assets	-0.7037***	-0.5788***
	(0.0935)	(0.0882)
llr_to_loans	-189.1***	-187.5***
	(4.283)	(4.400)
demand_dep_share	-0.3847	-0.2954
	(0.2643)	(0.2610)
time_dep_share	-1.917***	-1.513***
	(0.3913)	(0.3994)
roa	-24.36***	-21.10***
	(4.417)	(4.176)
post_basel × high_exposure	0.6539***	
	(0.0546)	
post_basel × exposure_cont		14.26***
		(1.967)
Fixed-effects		
id_rssd	Yes	Yes
qtr	Yes	Yes
Fit statistics		
Observations	371,507	371,507
R^2	0.21276	0.21554
Within R^2	0.04287	0.04625

Clustered (id_rssd) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

7.2 Variable Definitions

- **Loan Growth**: Log difference in total loans.
- **Exposure**: Pre-2010 equity-to-assets ratio (inverse).
- **Post**: Indicator for post-Basel III period (2013 onward).
- **LLR / Loans**: Loan loss reserves over total loans.
- **ROA**: Return on assets.